

SHUBHAM OJHA

Electrical Engineering (Dual Degree - B.Tech + M.Tech.)

Indian Institute of Technology Bombay

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Examination	University	Institute	Year	CPI/%
Graduation	IIT Bombay	IIT Bombay	2024	8.01
Intermediate	CBSE	Bhartiya Vidya Bhavan	2019	96.00%
Matriculation	CBSE	Bhartiya Vidya Bhavan	2017	95.00%

SCHOLASTIC ACHIEVEMENTS

- Achieved **All India Rank 533** out of 0.17 million candidates in **JEE Advanced** entrance exam (2019)
- Secured **99.94 percentile** in **JEE Main** entrance examination amongst 1.2 million candidates (2019)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer - Clarity (April 2025 – Present)

Building end-to-end ML infrastructure to enable optimal human-AI collaboration

- Engineered multiple **in-house ML models**, reducing reliance on third-party APIs and **cutting cost by 7×**
- Developed **Beam pipelines (Java)** for streaming and batch data, ensuring scalability and reliability
- Implemented a **GPU-backed Kubeflow** setup to automate and accelerate model training workflows
- Built high-throughput **Python workers (ARQ)** to handle large volumes of asynchronous file operations

Software Development Intern - Citrix | Summer Internship (May 2022 – July 2022)

Engineered a trace parser tool to convert log files into tabular format for streamlined analysis

- Revamped app via **ElectronJS** ensuring cross-platform for broader accessibility and streamlined upkeep
- Attained **10% speed boost** in processing and **50% RAM reduction** maximizing application performance
- Followed **Component Driven Development** approach to develop elements used in the application
- Designed **Web Workers** for parsing text files greater than **500MB** without affecting user experience

Software Development Intern - Newzera | Winter Internship (December 2020 – March 2021)

Collaborated on an AI powered web platform to reinvent the space of online journalism

- Created various pages and components in the primary web application using the **ReactJS** Library
- Implemented many GraphQL queries using the **Apollo GraphQL** using **MySQL** as the database system
- Achieved up to **95% coverage** by testing developed components using **Jest** and **Enzyme** frameworks

KEY PROJECTS

Lifted Trajectory Games | M.Tech. Project - Guide: Prof. Debraj Chakraborty (August 2023 – July 2024)

Developed strategies for multi-agent trajectory games

- Conducted comprehensive **literature review** on trajectory games and multi-agent interactions
- Utilized **Julia** programming for efficient real-time simulations with loop times under **10 milliseconds**
- Created **cost functions** for diverse game scenarios, including Cooperative Tag and Herding games

Metamask Snaps | Problem Statement from Inter IIT Tech Meet 11.0 (January 2023 – February 2023)

Expanded functionality of Metamask using their pre-release software Metamask Snaps

- Collaborated with 7 people to add **Account Abstraction** to popular web3 wallet Metamask using Snaps
- Defined several **JSON RPC** API endpoints for fast communication between the React frontend and chains
- Attained **90%+ functionality score**, constructing the UI and backend within a month's time frame

Greedy Cluster Margin | Course Project – Guide: Prof Ganesh Ramkrishnan (March 2023 – May 2023)

An efficient algorithm designed to maximize insights from training dataset while minimizing sample usage

- Pioneered new **active learning** algorithm using DECILE library for optimizing labelling efforts
- Improved efficiency by achieving **60x time reduction** in clustering algorithm while preserving accuracy
- Achieved a **30% label efficiency** boost compared to random sampling, on par with DECILE algorithms

Flip Food - Website | Self Project

(August 2020)

A mock e-commerce website featuring full user registration and integrated payment portals

- Developed the backend modules with a user management system using **Django** Web Framework
- Handled development of frontend user interface using HTML, CSS, **Bootstrap** and **JavaScript**

Emojify | Course Project

(March 2021)

Training AI to recognize emotions based on facial expressions

- Developed a Emotion Detection Classifier using **Tensorflow** on the FERN-2013 dataset
- Trained a Convolutional Neural Network with an accuracy of **82%** on training and **65%** on testing sets
- Implemented **Haar Cascade Classifier** to detect faces and replaced them with emojis using **OpenCV**

Traffic-MAN | Institute Technical Summer Project

(March 2020 – June 2020)

Pioneered a traffic light system addressing congestion, signal time wastage, and delays for emergency vehicles

- Collaborated with a team of three to build a **Django** application simulating a traffic light intersection
- Implemented **YOLOv3** for fast object extraction and **polynomial regression** algorithm for smart timing
- Enforced **Git** and **OOP** for organized and modular code, ensuring a systematic approach to traffic control

Anomaly Detection | Research Project – Guide: Prof. Nikhil Karamchandani

(Mar 2023 – May 2023)

Using EDA along with predictor detector system to detect erroneous data

- Programmed **1200+ bug free** lines for end-to-end anomaly detection pipeline in univariate time series
- Analysed and built report on dataset containing **200+ columns** using the **pandas** and **matplotlib** library
- Implemented **state-of-the-art** accumulator detector system using models like **Fourier**, **LSTMs**

POSITIONS OF RESPONSIBILITY

Web Manager | Unmesh Mashruwala Innovation Cell (UMIC)

(August 2021 – March 2022)

40+ member technical team with the primary focus on autonomous vehicles in the sky and the ground

- **Reduced operating costs** by shifting website deployed on DigitalOcean from Django to **Nuxt** Framework
- Decreased the server downtime by writing scripts that **automate** deployment of website on the server

Institute Technical Convener | Electronics and Robotics Club

(June 2020 – March 2021)

Part of two-tiered team of 80+ members organizing competitions, talks, and projects to promote tech culture

- Part of **17 membered core team** aimed to organize **20+ events**, competitions, talks (BLAHs), and hackathons and mentor **1200+** electronics and robotics enthusiasts across the institute
- Strengthened the club's social media presence increasing the reach by **300%** and likes by **50%**

EXTRACURRICULARS AND OTHER ACHIEVEMENTS

Social Volunteering	Volunteered for IITB's social body Abhyuday's Career Counselling Campaign in collaboration with NCC to make students aware of different career paths
NCC	• Participated in Republic Day Parade held on 26th Jan, 2020 at gymkhana ground • Among the 18 selected students for Captain's Training Camp out of 150 applicants
Technical Activities	• Completed Tinkering Bootcamp on Arduino+RPi learning via project and challenges • Achieved a rating of 1400+ on the popular competitive coding website - CodeChef • Finished IIT Bombay's Web Dev Course (Frontend, Backend) at Learners' Space 2020 • Mentored a team of 4 students building an application from scratch in ITSP 2021
Hobbies	• Passionate about K-Dramas, drawn to their storytelling, and character development • Casual gamer: Enjoy exploring Valorant tactics and Dark Souls adventures • Interested in playing football, singing, reading fiction books, and dancing